We can see that if $P(x,y)$ be any point on the unit circle in the first quadrant and $\angle POQ = \theta$

$$\sin\theta = \frac{PQ}{OP} = \frac{y}{1} = y \;;\quad \cos\theta = \frac{OQ}{OP} = \frac{x}{1} = x \;;\quad \tan\theta = \frac{PQ}{OQ} = \frac{y}{x}$$

When $\theta = 0°$, OP coincides with OA, where A is $(1,0)$ giving $x = 1,\ y = 0$.

We get thereby,

$$\sin 0° = 0 \;;\quad \text{cosec}\, 0° = \text{not defined (why?)}$$

$$\cos 0° = 1 \;;\quad \sec 0° = 1$$

$$\tan 0° = \frac{0}{1} = 0 \;;\quad \cot 0° = \text{not defined (why?)}$$

When $\theta = 90°$, OP coincides with OB, where B is $(0,1)$ giving $x = 0,\ y = 1$.

Hence,

$$\sin 90° = 1 \;;\quad \text{cosec}\, 90° = 1$$

$$\cos 90° = 0 \;;\quad \sec 90° = \text{not defined}$$

$$\tan 90° = \frac{1}{0} = \text{not defined} \;;\quad \cot 90° = 0$$

Let us summarise all the results in the table given below:

Trigonometric ratio \ θ	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	not defined
$\text{cosec}\,\theta$	not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	not defined
$\cot\theta$	not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

6 Trigonometry.indd 230 02/12/2022 21:47:35